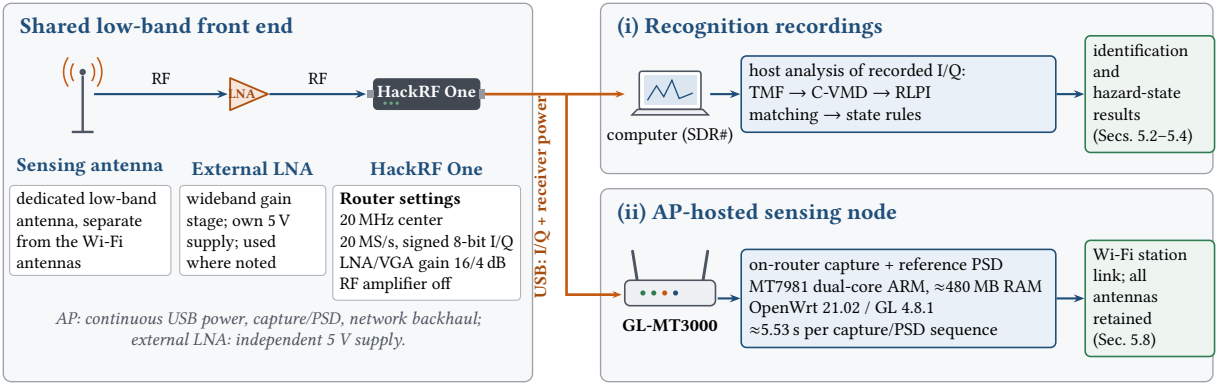

Shared low-band front end
RF
LNA
RF
HackRF One
Sensing antenna
dedicated low-band antenna, separate from the Wi-Fi antennas
External LNA
wideband gain stage; own 5 V supply; used where noted
HackRF One
Router settings
20 MHz center
20 MS/s, signed 8-bit I/Q
LNA/VGA gain 16/4 dB
RF amplifier off
AP: continuous USB power, capture/PSD, network backhaul;
external LNA: independent 5 V supply.
USB: I/Q + receiver power
(i) Recognition recordings
computer (SDR#)
host analysis of recorded I/Q:
TMF → C-VMD → RLPI
matching → state rules
identification and hazard-state results
(Secs. 5.2–5.4)
(ii) AP-hosted sensing node
GL-MT3000
on-router capture + reference PSD
MT7981 dual-core ARM, ≈480 MB RAM
OpenWrt 21.02 / GL 4.8.1
≈5.53 s per capture/PSD sequence
Wi-Fi station link; all antennas retained
(Sec. 5.8)